Sta301 by waqar finalterm

Arrange vu students

Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The parameter of the chi- square distribution is.....

Answer (Please select your correct option)

- ν — **correct answer solved by hadi**
- $\nu - 1$
- $\nu - 2$
- ν -p

Correct answer solved by HADI
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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The value of χ^2can never be :

Answer (Please select your correct option)

- Zero
- Less than 1
- Greater than 1
- Negative — **correct answer solved by hadi**

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

The probability of an event always lies between:

Answer (Please select your correct option)

- 0 and -∞
- -1 and +1
- -∞ and + -∞
- 0 and 1

correct answer
solved by hadi

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The number of parameters in a Poisson distribution is (are):

Answer (Please select your correct option)

- 1

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correct answer
solved by hadi

- 0
- 2
- 3

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

E(4X + 5) =_____

Answer (Please select your correct option)

- 16 E (X)
- 16 E (X) + 5
- 12 E (X)
- 4 E (X) + 5

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In a one-way ANOVA:

Answer (Please select your correct option)

- The interaction term has (c - 1)(n - 1) degrees of freedom
- An interaction term is given
- An interaction effect can be tested
- There is no interaction term

Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The degrees of freedom for a t-test with sample size 14 is:

Answer (Please select your correct option)

- 14
- 13

correct answer
solved by hadi

- 7
- 0

Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The degrees of freedom for a t-test with sample size 6 is:

Answer (Please select your correct option)

- 1
- 3
- 5

correct answer
solved by hadi

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- 7

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a characteristics of the normal distribution:

Answer (Please select your correct option)

- It is a skewed distribution
- It is bell-shaped

correct answer
solved by hadi

- It is not asymptotic
- It is leptokurtic

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution β_1 = ... :

if b2 then 3

Answer (Please select your correct option)

- 0

correct answer
solved by hadi

- 1
- 2
- 3

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution β_2 = ... :

Answer (Please select your correct option)

- 1
- 2
- 3

correct answer
solved by hadi

- 0

Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

- Repeated sampling
- Small sample
- Large sample

correct answer
solved by hadi

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- Large population

Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is a measure of absolute dispersion?

Answer (Please select your correct option)

- Skewness
- Mean Deviation

page 330

correct answer
solved by hadi

- Coefficient of variation
- Kurtosis

Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is mor appropriate to calculate the average speed?

Answer (Please select your correct option)

- Median
- Arithematic mean
- Harmonic mean

correct answer
solved by hadi

- Geometric mean

Question No : 15 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If $S.D(X) = 5$ then $S.D(\frac{2X+5}{2})$ = _____

Answer (Please select your correct option)

- 5
- 10
- 15
- 7.5

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For a particular data set the Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

Answer (Please select your correct option)

- Negatively skewed
- J-shaped
- Symmetrical
- Positively skewed

correct answer
solved by hadi

Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The total number of samples when sampling is done without replacement is equal to:

Answer (Please select your correct option)

- N^n

correct answer
solved by hadi

- C_n^N
- $\frac{N-n}{N-1}$
- 1

Question No : 18 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is correct option to find $P(X+Y \leq 1)$?

Answer (Please select your correct option)

- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

correct answer
solved by hadi

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Question No : 19 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

The conditional probability function $f(x \mid 1) =$ _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 20 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

If f (x, y) is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 21 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

- Hypothesis
- Lower limit
- Confidence interval

correct answer
solved by hadi not sure

- Probability limit

Made by: Waqar Siddhu

Question No : 22 of 52 **Marks: 1 (Budgeted Time 1 Min)**

By definition f(y | x) = _____

Answer (Please select your correct option)

- $f(y)$
- $f(x,y)$
- $\frac{f(x,y)}{h(x)}$

correct answer
solved by hadi

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- $\frac{f(x,y)}{h(y)}$

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is an unbiased estimator of population σ^2 ?

Answer (Please select your correct option)

- $\frac{\sum(x-\bar{x})^2}{n}$
- $\frac{\sum(x-\bar{x})^2}{n-2}$
- $\frac{\sum(x-\bar{x})^2}{n(n-1)}$
- $\frac{\sum(x-\bar{x})^2}{n-1}$

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 24 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If a significance level of 1% is used rather than 5%, the null hypothesis is:

Answer (Please select your correct option)

- More likely to be rejected
- Less likely to be rejected — correct answer solved by hadi
- Just as likely to be rejected
- None of the above

Made by: Waqar Siddhu

Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

- -2.58 and +2.58 — correct answer solved by hadi
- -2.33 and +2.33 — both one
- -1.645 and +1.645
- -1.96 and + 1.96 — correct answer solved by hadi

Made by: Waqar Siddhu

Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision — correct answer solved by hadi — idea
- No information regarding this

Made by: Waqar Siddhu

Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$

correct answer
solved by hadi

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- $P \geq 0.48$

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k

correct answer
solved by hadi

- 1/k
- k

Made by: Waqar Siddhu

Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

- Positively skewed

correct answer
solved by hadi

- Negatively skewed
- Uniformly distributed
- Normally distributed

Made by: Waqar Siddhu

Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

- Degree of freedom

correct answer
solved by hadi

- Critical value
- Frequency of data
- Probability

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The covariance of a random variable with itself is:

Answer (Please select your correct option)

- Zero
- One
- Its variance

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correct answer
solved by hadi

- Its correlation

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Hypergeometric probability distribution has :

Answer (Please select your correct option)

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

- Consistent
- Unbaised
- Sufficient
- Efficient

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Atmosphere pressure is the example of:

Answer (Please select your correct option)

- Constant
- Qualitative variable
- Quantitative variable
- None of the above

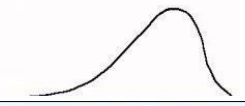
correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve
- Negatively skewed curve

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is not a type of frequency curve?

Answer (Please select your correct option)

- The symmetrical frequency curve
- The extremely skewed frequency curve
- The U-shaped frequency curve
- Frequency polygon

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

For positively skewed distribution
Mean.....Median.....Mode:

Answer (Please select your correct option)

- ≠
- =
- <
- >

correct answer
solved by hadi

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

What is 'f_m' in the formula of mode:

Answer (Please select your correct option)

- First Frequency
- Last Frequency
- Maximum Frequency
- Minimum Frequency

correct answer
solved by hadi

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is the measure of central tendency:

Answer (Please select your correct option)

- Variation of the distribution
- Average of the distribution

correct answer
solved by hadi

- Scatterness of the distribution
- Dispersion of the distribution

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

In a set of 10 values all the values are 5, what will be the P_{50}?

Answer (Please select your correct option)

- 2
- 5

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correct answer
solved by hadi

because every value are 5

- 10
- 20

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the name of methods/techniques that are used to represent the quantitative discrete data.

Answer (Please click here to Add Answer)

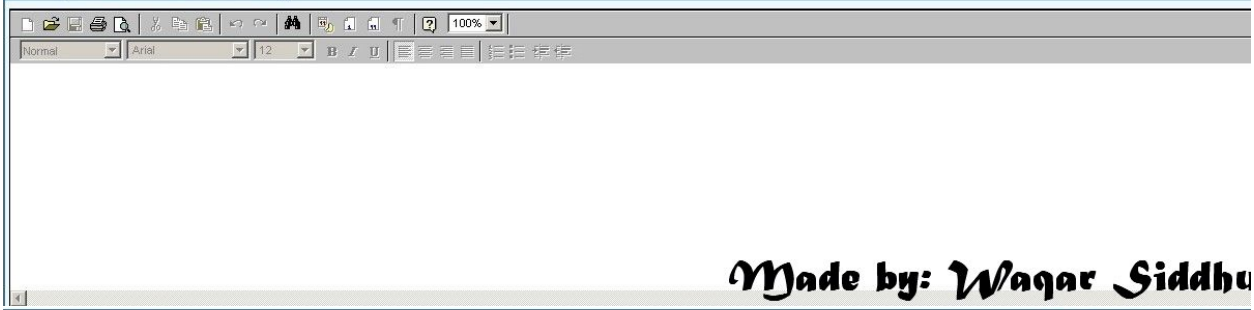


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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

A coin is tossed 900 times and heads appear 490 times. State the null and alternative hypotheses to show that the coin is unbiased.

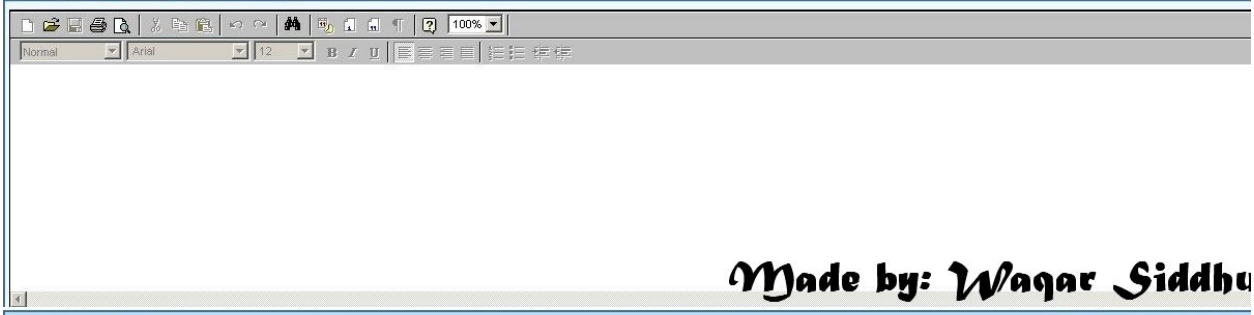
Answer (Please click here to Add Answer)



Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the names of two types of experimental designs.

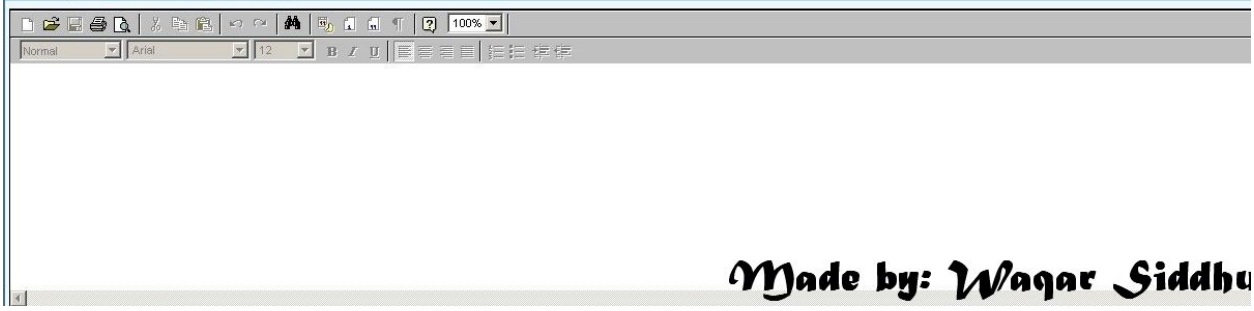
Answer (Please click here to Add Answer)



Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the mean and variance of binomial distribution?

Answer (Please click here to Add Answer)



Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the probability of drawing a white ball from a bag containing 4 red, 8 black and 3 white balls.

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down the properties of sampling distribution of proportion $\hat{p}$, when sampling is performed *without* replacement.

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If $X = 255$, $n = 500$, $p_0 = 0.60$ then find the z-test statistic for proportion.

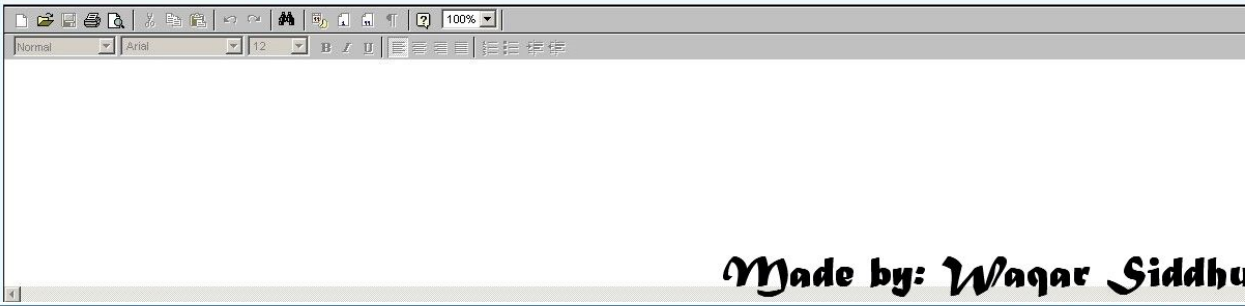
Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the impact of degrees of freedom on chi- square distribution?

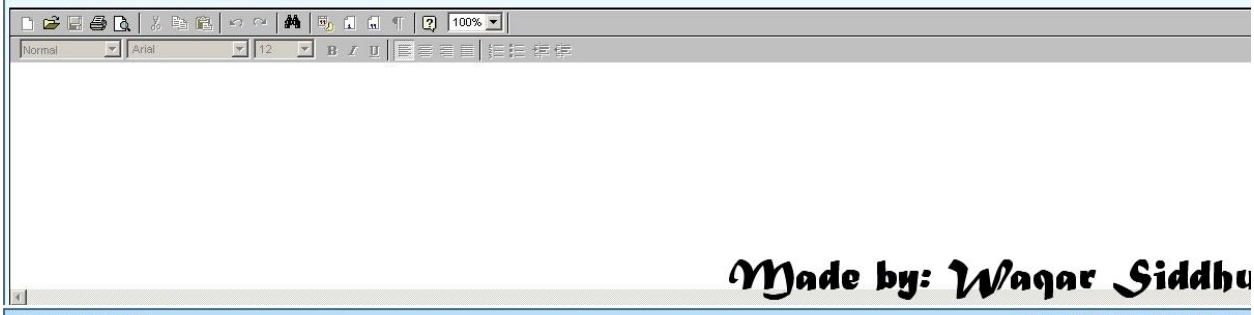
Answer (Please click here to Add Answer)



Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Find quartile deviation for the data given below:
18, 53, 45, 28, 39, 29, 23, 40 and 21

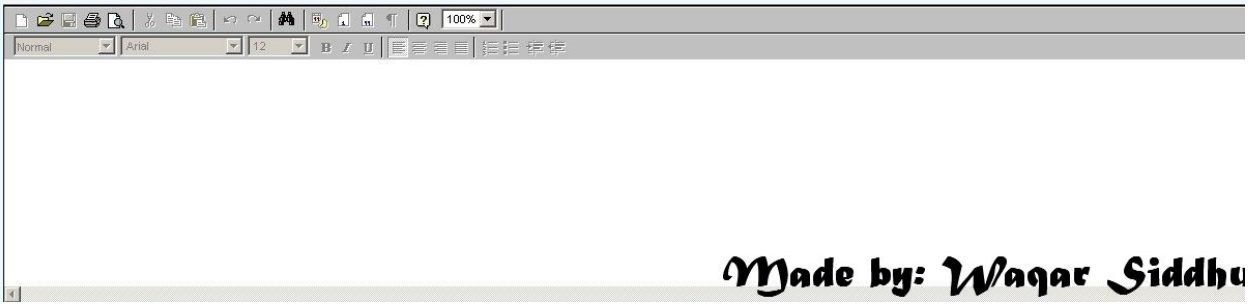
Answer (Please click here to Add Answer)



Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

A population consists of N=5 values 1, 2, 3, 5, 6.A sample size of n=3 is selected from the population without replacement, calculate sampling distribution of sample proportions for even numbers.

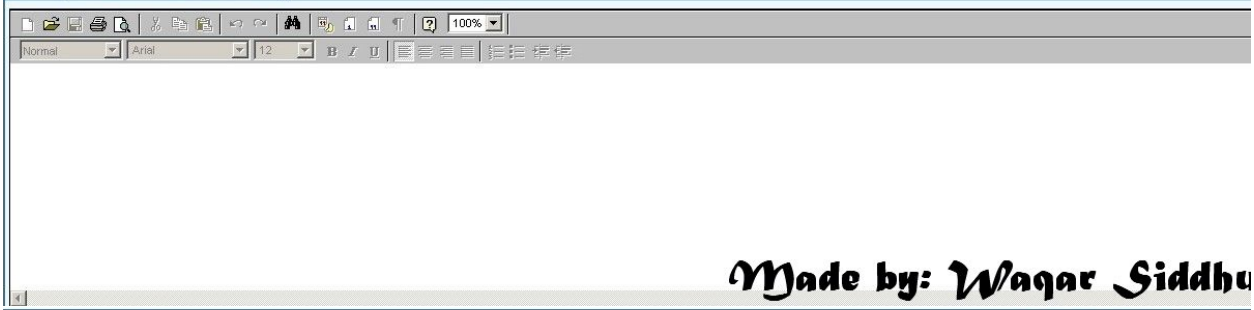
Answer (Please click here to Add Answer)



Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Write down the testing procedure in case of goodness of fit test.

Answer (Please click here to Add Answer)



Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

If we have n=634 and $\hat{p}$ =0.459, where $Z_{\frac{0.01}{2}}$=2.58, then find the 99% confidence interval for population proportion.

Answer (Please click here to Add Answer)



Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

P (A or B) =P (A) +P (B), then A and B are:

Answer (Please select your correct option)

- Mutually exclusive events — correct answer solved by hadi
- Independent events
- Exhaustive events
- Equally likely events

Correct answer solved by HADI
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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

First moment about origin is always equals to:

Answer (Please select your correct option)

- Mean
- Variance
- Standard Deviation
- Zero

correct answer
solved by hadi

Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

E(4X + 5) =_____

Answer (Please select your correct option)

- 16 E (X)
- 16 E (X) + 5
- 12 E (X)
- 4 E (X) + 5

correct answer solved by hadi

Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When two coins are tossed the probability of at least one head is:

Answer (Please select your correct option)

- 1/4
- 3/4
- 2/4
- 4/4

correct answer
solved by hadi

Correct answer solved by HADI
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Made by: Waqar Siddhu

Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If σ^2 is unknown, then we use Z-test if the sample size is:

Answer (Please select your correct option)

- $n \geq 30$
- n<30

correct answer
solved by hadi

- n=25
- n=20

Made by: Waqar Siddhu

Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When a coin is tossed 3 times, the probability of 3 tails is:

Answer (Please select your correct option)

- 1/8
- 2/4
- 3/8

correct answer
solved by hadi

- 2/8

Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The F-distribution has ..... parameter.

in t-distr has one

Answer (Please select your correct option)

- One
- No
- Two

correct answer
solved by hadi

- Three

Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following provides the basis for hypothesis testing?

Answer (Please select your correct option)

- Null hypothesis
- Alternative hypothesis
- Critical value
- Test-statistic

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

Answer (Please select your correct option)

- $H_1 : \mu = 100$
- $H_1 : \mu > 100$
- $H_1 : \mu < 100$
- $H_1 : \mu \neq 100$

correct answer
solved by hadi

Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

- $H_0 : \mu = 100$

correct answer
solved by hadi

- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

By definition f(x_i | y_j) = _____

Answer (Please select your correct option)

- $\frac{f(x_i, y_j)}{h(y_j)}$

correct answer
solved by hadi

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- $\frac{f(x_i, y_j)}{h(x_i)}$
- $f(x_i, y_j)$
- $f(y_j)$

Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The test statistic used in analysis of variance procedure follow the :

Answer (Please select your correct option)

- χ^2 -distribution.
- T-distribution.
- Z-distribution.
- F-distribution.

correct answer
solved by hadi

Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following is the most common example of a situation for which the main parameter of interest is a population proportion?

Answer (Please select your correct option)

- An observational study
- A normal experiment
-
- A binomial experiment

correct answer
solved by hadi

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

- Unbiasedness
- Efficiency

correct answer
solved by hadi

- Consistency
- Sufficiency

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

- Unbiasedness
- Efficiency

correct answer
solved by hadi

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- Consistency
- Sufficiency

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following can never be taken as the probability of an event?

Answer (Please select your correct option)

- 1
- 0
- 0.5
- -0.5

correct answer
solved by hadi

Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A set is any well-defined collection of

Answer (Please select your correct option)

- Positive Objects
- Negative Objects
- Same Objects
- Distinct Objects

correct answer
solved by hadi

Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Measure of dispersion is used to calculate the:

Answer (Please select your correct option)

- Central value
- Highest value
- Lowest value
- Scattered value

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 18 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If X and Y are independent variables then Var (X-Y) = _____

Answer (Please select your correct option)

- Var(X) - Var(Y)

correct answer
solved by hadi

- Var(X) + Var(Y)
- Var(X+Y)
- Var(X) ×Var(Y)

Made by: Waqar Siddhu

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

How the standard error is decreased :

Answer (Please select your correct option)

- By decreasing the sample size
- By decreasing the mean
- By increasing the standard deviation
- By increasing the sample size

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The total number of samples when sampling is done with replacement is equal to:

Answer (Please select your correct option)

- N^n

correct answer
solved by hadi

Correct answer solved by HADI
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- C_n^N
- $\frac{N-n}{N-1}$
- 1

Made by: Waqar Siddhu

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

- A population distribution
- A sample distribution
- A sampling distribution

correct answer
solved by hadi

- A parameter distribution

Made by: Waqar Siddhu

Question No : 22 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The conditional probability function $f(x|1)$ = _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$ — correct answer solved by hadi
- $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

- -2.58 and +2.58
- -2.33 and +2.33
- -1.645 and +1.645
- -1.96 and + 1.96 — correct answer solved by hadi

Made by: Waqar Siddhu

Question No : 24 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

- Type I error
- Type II error — correct answer solved by hadi

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- Correct decision
- No information regarding this

Made by: Waqar Siddhu

Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

Answer (Please select your correct option)

- Confidence interval — correct answer, solved by hadi
- Critical region
- Acceptance region
- Critical value

Made by: Waqar Siddhu

Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$ — correct answer, solved by hadi
- $P \geq 0.48$

Made by: Waqar Siddhu

Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

Answer (Please select your correct option)

- Z-statistic
- T-statistic
- χ^2-statistics — but not sure due to concept — correct answer, solved by hadi
- F-statistics

Made by: Waqar Siddhu

Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

To find the confidence interval for the ratio of two variances, we use

Answer (Please select your correct option)

- F-Distribution — correct answer solved by hadi — Correct answer solved by HADI Email : usmanraj20@gmail.com Cell : 03228043306
- Z-Distribution
- Chi-square-Distribution
- t-Distribution

Made by: Waqar Siddhu

Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The Chi- Square distribution is continuous distribution ranging from:

Answer (Please select your correct option)

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- $0 \leq \chi^2 \leq \infty$ — correct answer solved by hadi

Made by: Waqar Siddhu

Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In Statistics, we have MSE which is abbreviation of

Answer (Please select your correct option)

- Mean square error — correct answer solved by hadi
- Measured square error
- Medical screening exam
- Major sampling error

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

In a binomial experiment the total number of trials are:

Answer (Please select your correct option)

- Fixed in advance

correct answer
solved by hadi

Correct answer solved by HADI
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- Changeable according to situation
- Unpredictable
- Not independent

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The lottery tickets issued for the purpose of money-making follows a:

Answer (Please select your correct option)

- Normal distribution
- Discrete uniform distribution

correct answer
solved by hadi

- Binomial distribution
- Hypergeometric distribution

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following value could not represent a coefficient of correlation?

Answer (Please select your correct option)

- r = 0.99
- r = 1.09
- r = -0.73
- r = -1

Made by: Waqar Siddhu

Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In a one way ANOVA test there are 5 observations in each of three treatments. The degrees of freedom for the treatments is:

Answer (Please select your correct option)

- 5
- 3
- 1
- 2

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) =

Answer (Please select your correct option)

- 0.05
- 0.80

due to formula

correct answer
solved by hadi

- 0.95
- 0.75

Made by: Waqar Siddhu

Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

Answer (Please select your correct option)

- 0,1,2,3

correct answer
solved by hadi

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- 1,3,3,1
- 1, 2, 3
- 3, 2

Made by: Waqar Siddhu

Question No : 37 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

When f(x) is continuous probability function, then $P(X = 2)$ is:

Answer (Please select your correct option)

- 1
- 0.5
- 0 — correct answer solved by hadi
- 0.25

Made by: Waqar Siddhu

Question No : 38 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

Atmosphere pressure is the example of:

Answer (Please select your correct option)

- Constant
- Qualitative variable
- Quantitative variable — correct answer solved by hadi

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- None of the above

Made by: Waqar Siddhu

Question No : 39 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

Which of the following scale has true zero point?

Answer (Please select your correct option)

- Ratio Scale — correct answer solved by hadi
- Interval scale
- Nominal scale
- Ordinal scale

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given seires:

Answer (Please select your correct option)

- 4
- 3
  3
- 2
  3
  2
- 1
  3
  2*

correct answer
solved by hadi

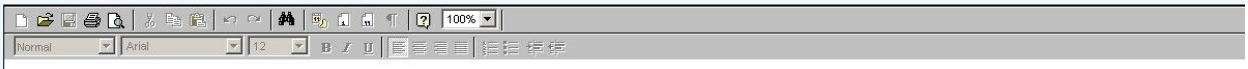
Correct answer solved by HADI
Email : us[illegible]@gmail.co[illegible]
Cell : 03[illegible]

Made by: Waqar Siddhu

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Describe the Poisson distribution as the limiting form of the binomial distribution.

Answer (Please click here to Add Answer)

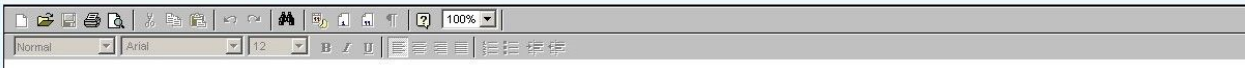


Made by: Waqar Siddhu

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Define Disjoint Sets.

Answer (Please click here to Add Answer)

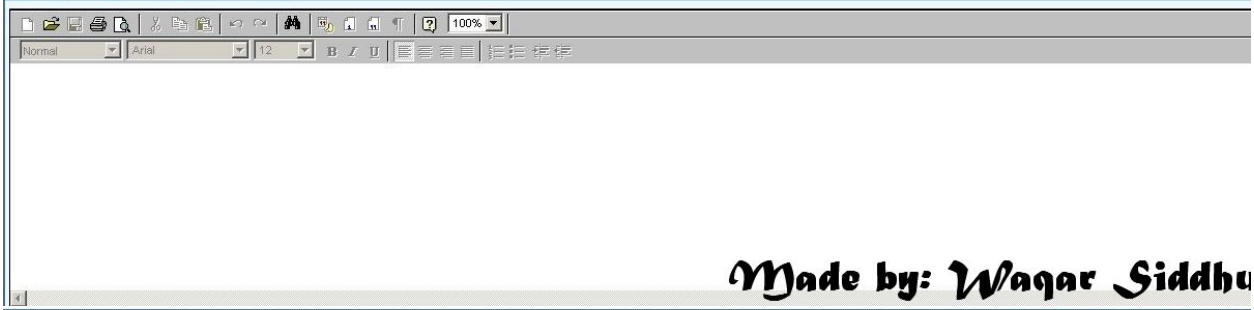


Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is acceptance region?

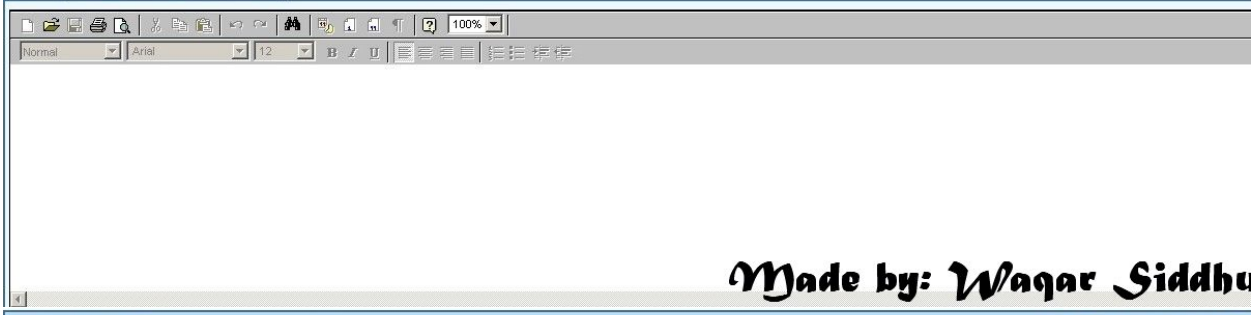
Answer (Please click here to Add Answer)



Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

If there are K treatments and R rows in a Randomized Complete Block Ddesign then calculate the total number of experimental units used.

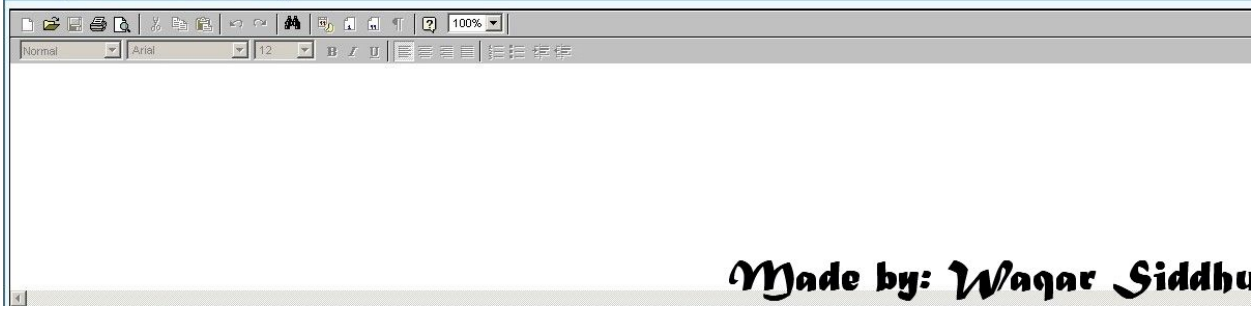
Answer (Please click here to Add Answer)



Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

In how many ways a three-person committee can be formed from a group of ten persons? (Use the formula)

Answer (Please click here to Add Answer)



Question No : 46 of 52 **Marks: 3 (Budgeted Time 6 Min)**

From the given data, calculate mean and standard deviation of sampling distribution of mean if the sampling is done with replacement.

$N = 120, n = 64, \mu = 50, \sigma = 2$

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Construct 90% confidence interval for the difference in means $\mu_1 - \mu_2$ in case of paired observations, where

$\bar{d} = 1.8$, $s_d = 1.32$, $t_{0.05(9)} = 1.833$

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

For a data X : $N(\mu, \sigma^2)$. Two unbiased estimators T_1 and T_2 have following variances

$V(T_1) = \frac{11\sigma^2}{25}$

$V(T_2) = \frac{9\sigma^2}{25}$

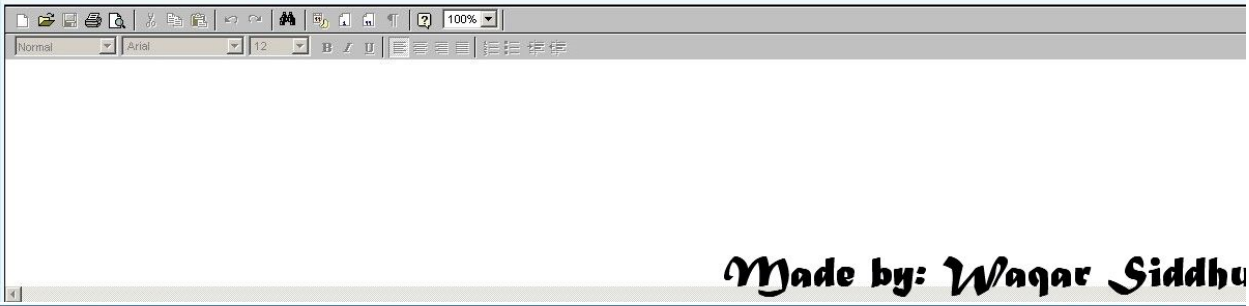
Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A man travels on car from Lahore to karachi on Motor way in 8 stages of equal intervals. The speed of the car in the various stages was observed to be 110, 116, 120, 114, 115,112, 120, 117 kilometers per hour. Find the average speed at which the car travels.

Answer (Please click here to Add Answer)

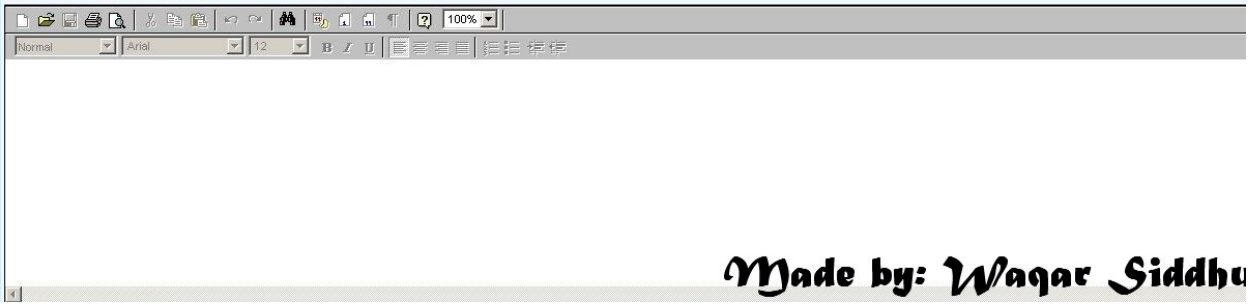


Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A random sample of size three is drawn without replacement from the population consisting of four numbers 4, 5, 5, 7. Sampling distribution of sample means is calculated as below;

Sample Means $(\bar{X})$	$f(\bar{X})$
14/3	1/4
16/3	2/4
17/3	1/4

Answer (Please click here to Add Answer)



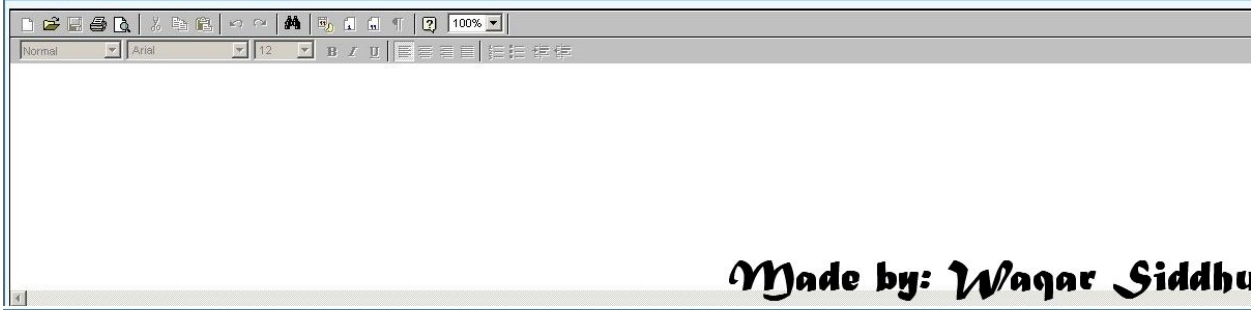
Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

If $n = 1150, x = 450, p = 0.39, H_0 : p_0 = 0.5, and\ \alpha = 0.05$

Test the stated hypothesis.

(*Use table value of* $z = \pm 1.96$)

Answer (Please click here to Add Answer)



Made by: Waqar Siddhu

Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A personal manager is interested in trying to determine whether absenteeism is greater on some specific day of the week or not? His records for the past year show the following sample distributions.

Day of the week:	Monday	Tuesday	Wednesday	Thursday	Friday
No. of absentees:	66	57	54	48	75

Test goodness of fit.

Where test statistic $chi - square = \sum \frac{(o_i - e_i)^2}{e_i} = 7.50$

Answer (Please click here to Add Answer)



Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

Answer (Please select your correct option)

- Mutually exclusive
- Equally likely

correct answer
solved by hadi

Correct answer solved by HADI
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- Not mutually exclusive
- Exhaustive

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

Answer (Please select your correct option)

- Mutually exclusive
- Equally likely

correct answer
solved by hadi

- Not mutually exclusive
- Exhaustive

Start Time: 9:48 AM

An expected value of a random variable is equal to:

Answer (Please select your correct option)

- Variance
- Mean

correct answer
solved by hadi

Correct answer solved by HADI
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- Standard deviation
- Covariance

Start Time: 9:48 AM

When f(x) is continuous probability function, then P(X = 1) is:

Answer (Please select your correct option)

- 1
- ∞
- $-\infty$
- 0

correct answer
solved by hadi

Made by: Waqar Siddhu

The sum of deviations is zero, when deviations are taken from:

Answer (Please select your correct option)

- Mean

correct answer
solved by hadi

- Median
- Mode
- H.M

Made by: Waqar Siddhu

The distribution function F(x) is equal to

Answer (Please select your correct option)

- $P(X = x)$
- $P(X \leq x)$

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correct answer
solved by hadi

- $P(X \geq x)$
- $P(X > x)$

In a one-way ANOVA:

Answer (Please select your correct option)

- The interaction term has (c - 1)(n - 1) degrees of freedom
- An interaction term is given
- An interaction effect can be tested
- There is no interaction term

Start Time: 9:48 AM

The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

- n-1 **correct answer solved by hadi**
- n+1
- n-2
- n+2

Start Time: 9:48 AM

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

- $H_0 : \mu = 100$ **correct answer solved by hadi**
- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

Correct answer solved by HADI
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Start Time: 9:48 AM

When c is a constant, then E(c) is:

Answer (Please select your correct option)

- 0
- 1
- c — **correct answer**
  solved by hadi
- -c

Start Time: 9:48 AM

The combined distribution of more than two random variables is:

Answer (Please select your correct option)

- Univariate distribution
- Joint distribution — **correct answer**
  solved by hadi
- Marginal distribution
- Bivariate distribution

Start Time: 9:48 AM

The test statistic used in analysis of variance procedure follow the :

Answer (Please select your correct option)

- χ^2 -distribution.
- T-distribution.
- Z-distribution.
- F-distribution. — **correct answer**
  solved by hadi

Correct answer
solved by HADI
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Start Time: 9:48 AM

In normal distribution β_2 = ... :

Answer (Please select your correct option)

- 1
- 2
- 3

correct answer
solved by hadi

- 0

Start Time: 9:48 AM

In normal distribution, the quartile deviation Q.D = ... :

Answer (Please select your correct option)

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

correct answer
solved by hadi

Start Time: 9:48 AM

In a symmetrical distribution, the coefficient of skewness is equal to :

Answer (Please select your correct option)

- -1
- +1
- 0

correct answer
solved by hadi

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- 2

Start Time: 9:48 AM

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

- Geometric Mean
- Harmonic Mean — **correct answer solved by hadi**
- Mode
- Median

Start Time: 9:48 AM

Which measure of dispersion is used to compare the variation of two data sets?

Answer (Please select your correct option)

- Coefficient of variation — **correct answer solved by hadi**
- Coefficient of comparison
- Mean deviation
- Standard deviation

Made by: Waqar Siddhu

Start Time: 9:48 AM

If $S.D(X) = 5$ then $S.D(\frac{2X+5}{2})$ = _____

Answer (Please select your correct option)

- 5
- 10
- 15
- 7.5 — **correct answer solved by hadi**

Made by: Waqar Siddhu

Start Time: 9:48 AM

The deviation of a distribution from symmetry is called:

Answer (Please select your correct option)

- Kurtosis
- Skewness

correct answer solved by hadi

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- Dispersion
- Flatness

Made by: Waqar Siddhu

Start Time: 9:48 AM

The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$

correct answer solved by hadi

- $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

Start Time: 9:48 AM

Which one is the correct formula to find the desired sample size?

Answer (Please select your correct option)

- $n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{e} \right)^2$

correct answer solved by hadi

- $n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{\sigma}}{e} \right)^2$
- $n = \left(\frac{Z_{\alpha/2} \cdot \bar{X}}{e} \right)^2$
- $n = \frac{Z_{\alpha/2} \cdot \sigma}{e}$

Made by: Waqar Siddhu

Start Time: 9:48 AM

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

- Type I error
- Type II error

correct answer
solved by hadi

Correct answer solved by HADI
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- Correct decision
- No information regarding this

Made by: Waqar Siddhu

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision

correct answer
solved by hadi

by concept

- No information regarding this

Made by: Waqar Siddhu

Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

- 0

correct answer
solved by hadi

- 1
- 99
- 100

Made by: Waqar Siddhu

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

- $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 2.58\sigma_{\bar{x}}$ **correct answer solved by hadi**
- $\mu_{\bar{x}} \pm \sigma_{\bar{x}}$

Made by: Waqar Siddhu

Start Time: 9:48 AM

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k **correct answer solved by hadi** Correct answer solved by HADI Email : usmanraj20@gmail.com Cell : 03228043306
- 1/k
- k

Made by: Waqar Siddhu

Start Time: 9:48 AM

Mean of the F-distribution is possible only, when

Answer (Please select your correct option)

- $v_1 > 2$
- $v_2 > 2$ **correct answer solved by hadi**
- $v_1 < 2$
- $v_2 < 2$

Made by: Waqar Siddhu

Start Time: 9:48 AM

In Statistics, we have MSE which is abbreviation of:

Answer (Please select your correct option)

- Mean square error — **correct answer solved by hadi**
- Measured square error
- Medical screening exam
- Major sampling error

Made by: Waqar Siddhu

Start Time: 9:48 AM

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

- Positively skewed — **correct answer solved by hadi** Correct answer solved by HADI Email : usmanraj20@gmail.com Cell : 03228043306
- Negatively skewed
- Uniformly distributed
- Normally distributed

Made by: Waqar Siddhu

Start Time: 9:48 AM

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution — **correct answer solved by hadi**

Made by: Waqar Siddhu

Start Time: 9:48 AM

If X and Y are independent variables, then E (XY) is:

Answer (Please select your correct option)

- E(XX)
- E(X).E(Y)

correct answer
solved by hadi

- X.E(Y)
- Y.E(X)

Made by: Waqar Siddhu

Start Time: 9:48 AM

What are the number of ways in which four books can be arranged on a shelf?

Answer (Please select your correct option)

- 4
- 6
- 12
- 24

correct answer
solved by hadi

Made by: Waqar Siddhu

Start Time: 9:48 AM

The parameters of the binomial distribution b(x; n, p) are:

Answer (Please select your correct option)

- x & n
- x & p
- n & p

correct answer
solved by hadi

Correct answer solved by HADI
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- x, n & p

Made by: Waqar Siddhu

Start Time: 9:48 AM

Hypergeometric probability distribution has :

Answer (Please select your correct option)

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

correct answer
solved by hadi

Made by: Waqar Siddhu

Start Time: 9:48 AM

The variance of the hypergeometric probability distribution is:

Answer (Please select your correct option)

- $\sigma^2 = n\frac{k}{N}$
- $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$,

correct answer
solved by hadi

- $\sigma^2 = npq$
- $\sigma^2 = N\frac{n}{K}$

Made by: Waqar Siddhu

Start Time: 9:48 AM

When f(x) is continuous probability function for $1 < X < 5$, then $P(X < 1)$ is:

Answer (Please select your correct option)

- 0

correct answer
solved by hadi

but check

- 0.25
- 0.5
- 1

Made by: Waqar Siddhu

Start Time: 9:48 AM

For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

- Unbaised
- Sufficient
- Efficient

correct answer
solved by hadi

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- Consistent

Made by: Waqar Siddhu

Start Time: 9:48 AM

If an estimator gets closer to the population parameter by increasing sample size then it is known as:

Answer (Please select your correct option)

- Consistent estimator
- Sufficient estimator
- Efficient estimator
- Unbiased estimator

Made by: Waqar Siddhu

Start Time: 9:48 AM

Which of the following comes first to make frequency distribution.

Answer (Please select your correct option)

- Number of Groups
- Class interval
- Range

correct answer
solved by hadi

- Tally marks

Made by: Waqar Siddhu

Start Time: 9:48 AM

What curve shape would you expect for the distribution of death rates of population of all age groups?

Answer (Please select your correct option)

- Symmetrical curve
- Skewed to the right
- Skewed to the left
- U shape curve

correct answer
solved by hadi

Made by: Waqar Siddhu

Start Time: 9:48 AM

Which one is the measure of central tendency:

Answer (Please select your correct option)

- Variation of the distribution
- Average of the distribution
- Scatterness of the distribution
- Dispersion of the distribution

Correct answer solved by HADI
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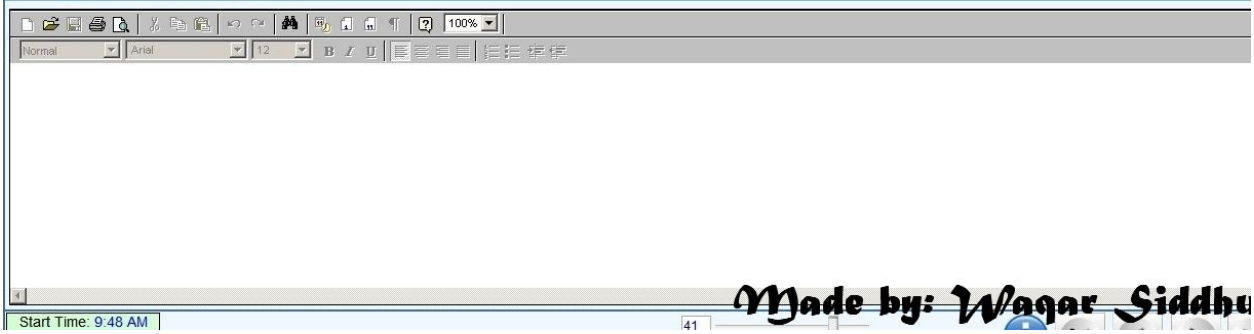
correct answer
solved by hadi

Made by: Waqar Siddhu

Start Time: 9:48 AM

What does quartile deviation measure in a distribution?

Answer (Please click here to Add Answer)

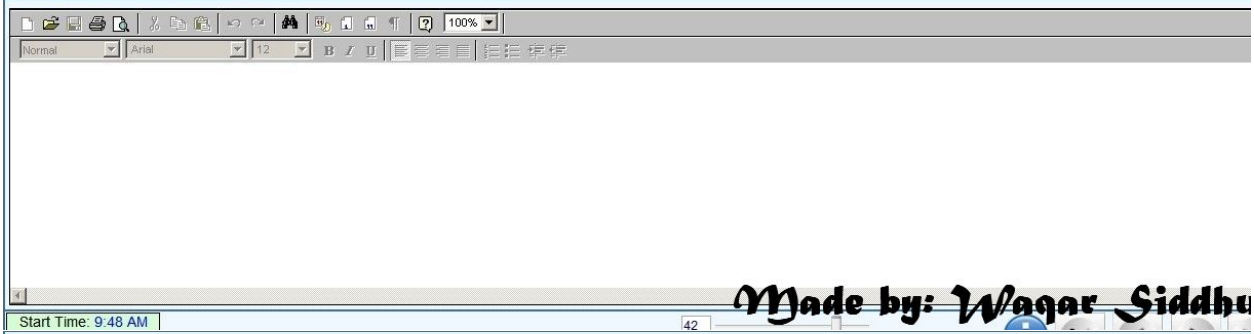


Made by: Waqar Siddhu

Start Time: 9:48 AM

In a random sample of 1000 women in a locality, 224 women said that they use Aerial washing powder. What is the point estimator and point estimate of the proportion of the women who use Aerial washing powder?

Answer (Please click here to Add Answer)



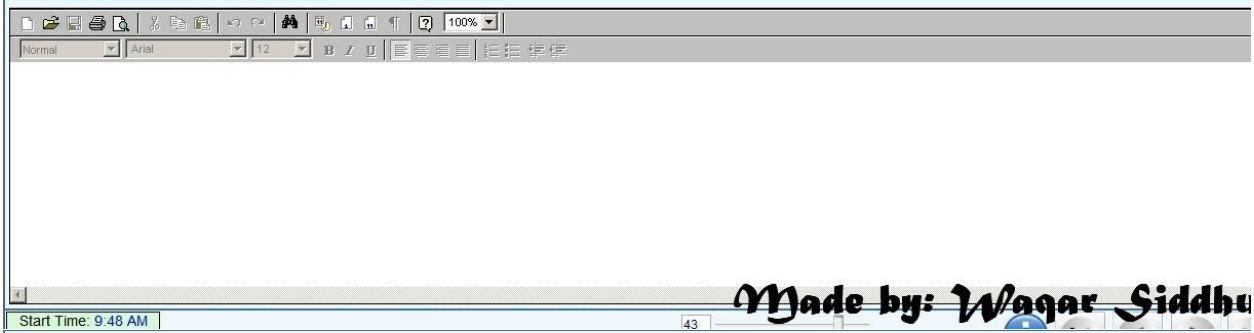
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Start Time: 9:48 AM

42

What is the impact of level of confidence (1-α) on the value of $Z_{\alpha/2}$?

Answer (Please click here to Add Answer)



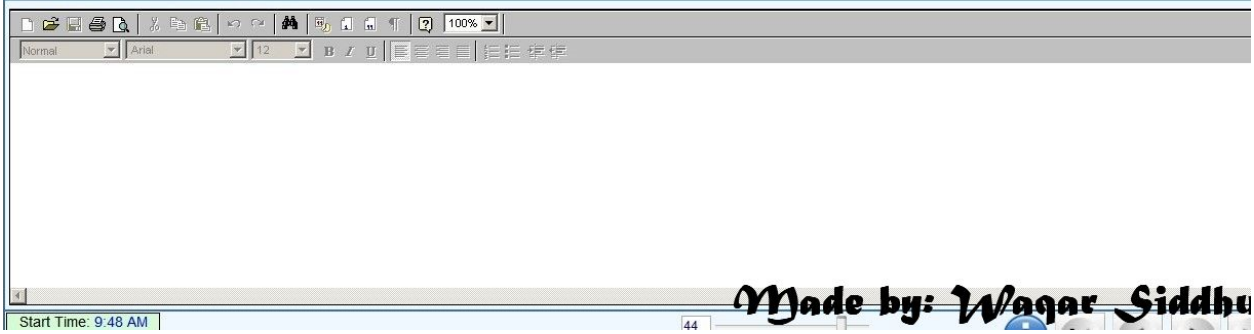
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Start Time: 9:48 AM

43

Formulate the hypothesis in case of goodness of fit test.

Answer (Please click here to Add Answer)



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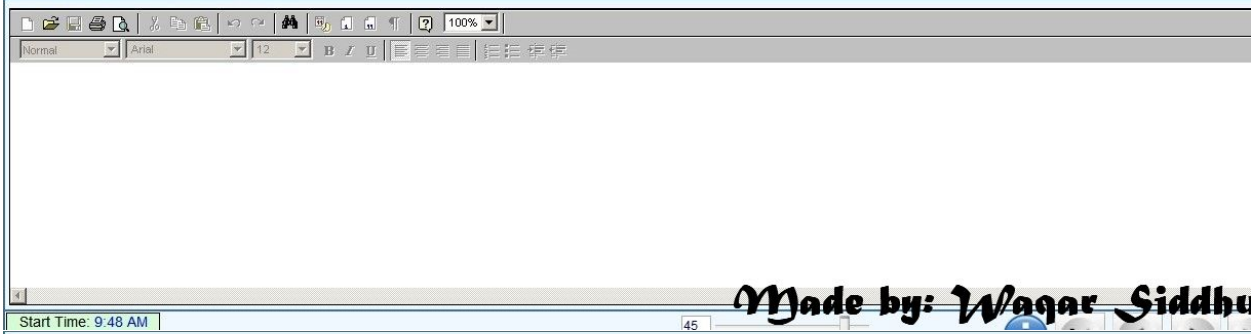
Start Time: 9:48 AM

44

Why the median is suitable average for the below data set? Explain.

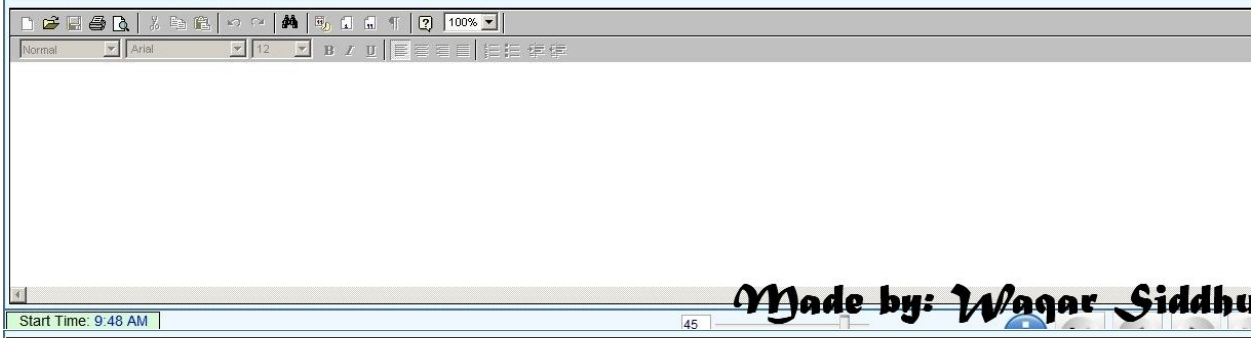
Monthly income (in rupees)	No. of workers
Less than 2000/-	100
2000/-to 2999/-	300

Answer (Please click here to Add Answer)



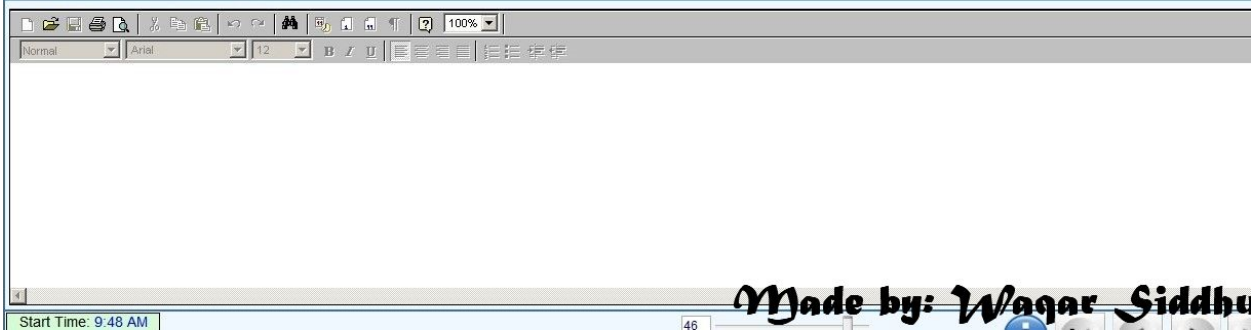
(in rupees)	
Less than 2000/-	100
2000/-to 2999/-	300
3000/- to 3999/-	500
4000/- to 4999/-	250
5000/- and above	50

Answer (Please click here to Add Answer)



If Z is a standard normal random variable with mean 0 and variance 1, then find the Lower quartile.

Answer (Please click here to Add Answer)

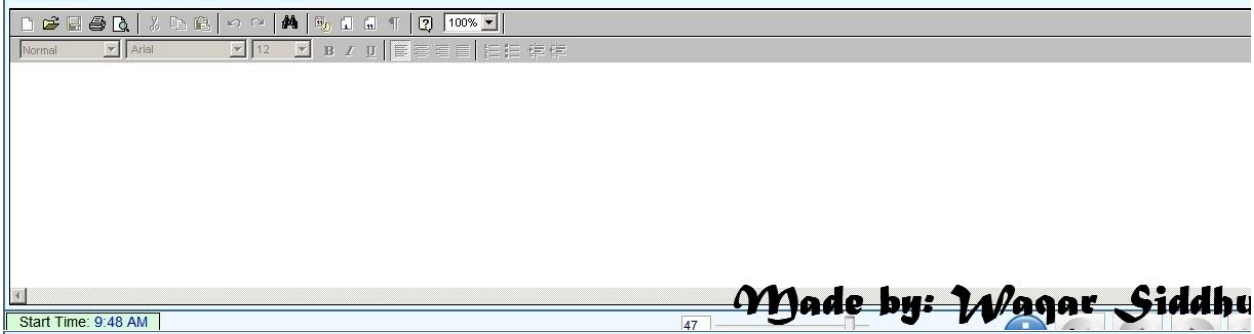


Write down critical region for the following hypothesis at 5% level of significance.

$H_0: \mu = 75$

$H_1: \mu < 75$

Answer (Please click here to Add Answer)



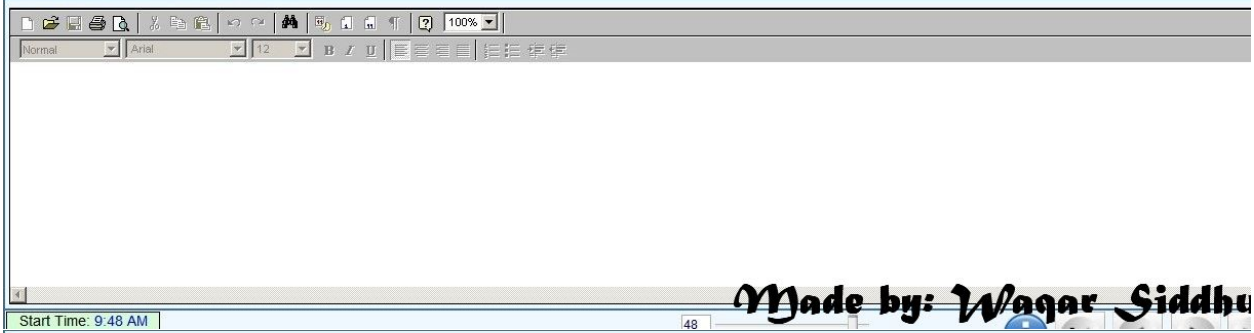
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Start Time: 9:48 AM

47

The manager of a bottling plant is anxious to reduce the variability in net weight of fruit bottled. Over a long period, the standard deviation has been 15.2 gm. A new machine is introduced and the ne
weights (in grams) in 10 randomly selected bottles (all of the same nominal weight) are 987, 966, 955, 977, 981, 967, 975, 980, 953, and 972.
State null and alternative hypothesis that machine has a better performance? Also write down test-statistics about the hypothesis.

Answer (Please click here to Add Answer)



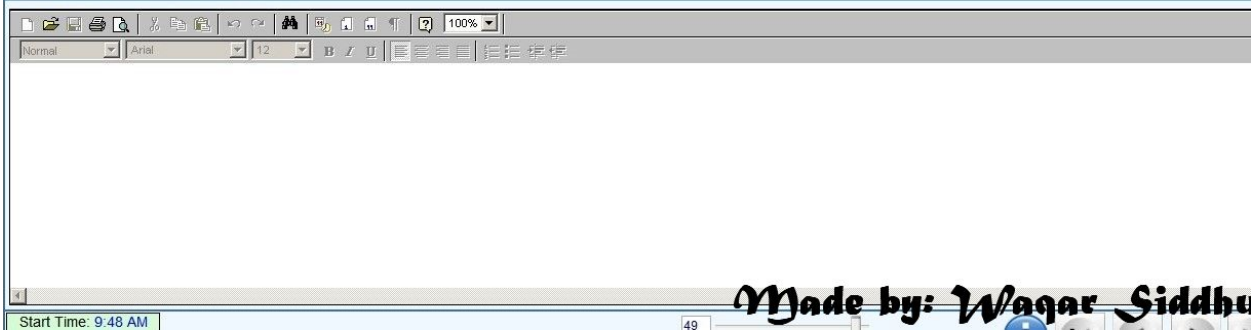
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Start Time: 9:48 AM

48

Find the coefficient of standard deviation from the following data:
Life in Hours (X): 130, 150, 180, 250, 345

Answer (Please click here to Add Answer)



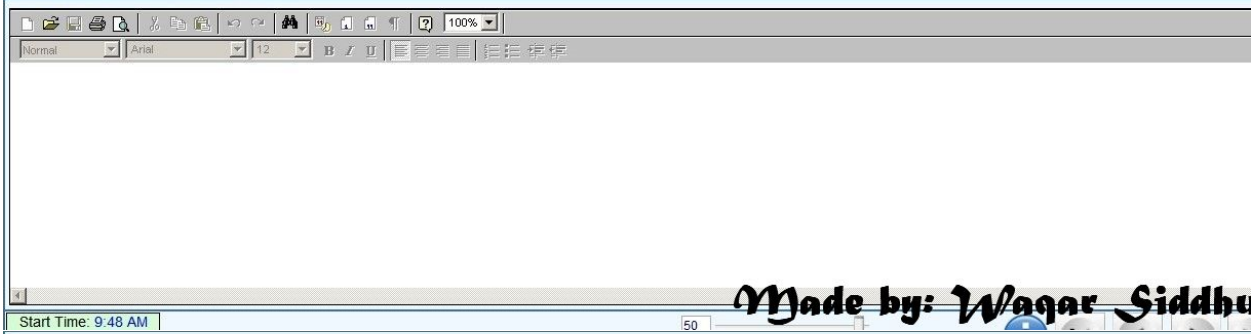
Made by: Waqar Siddhu

Start Time: 9:48 AM

49

What is the probability that a poker hand of 5 cards contain (i) exactly 2 aces (ii) exactly 1 ace?
(Use hypergeometric distribution)

Answer (Please click here to Add Answer)



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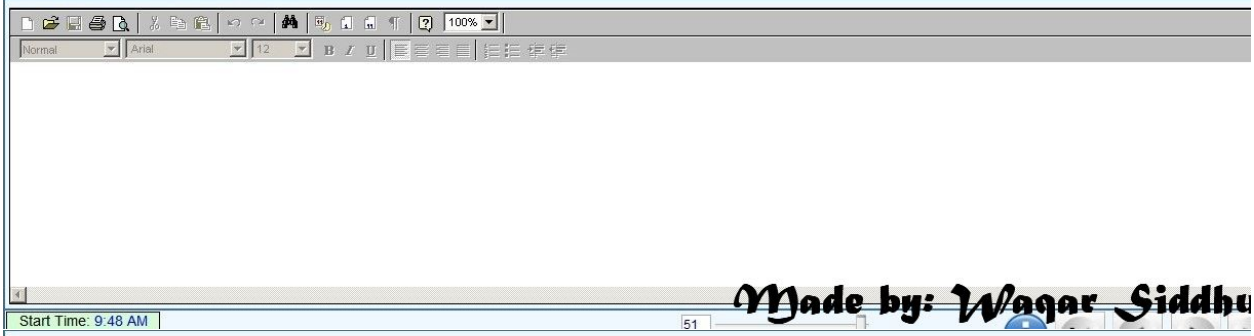
Start Time: 9:48 AM

50

If n=22, $\sum\left(d-\bar{d}\right)^2 = 270, \bar{d} = 4$

Then, using a 0.05 level of significance level, test the hypothesis $H_0 : \mu_d = 0$ against $H_1 : \mu_d \neq 0$

Answer (Please click here to Add Answer)



Made by: Waqar Siddhu

Start Time: 9:48 AM

51

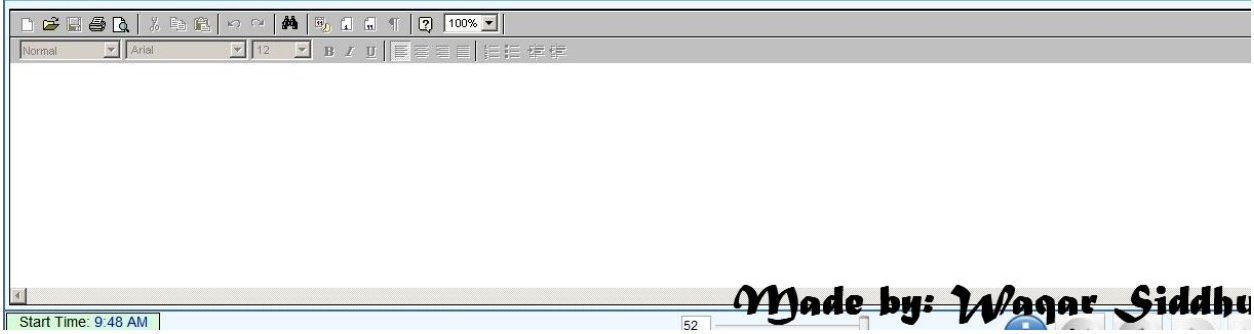
A company launched new layout of its website. After a survey, 62 of 115 visitors liked the new layout while 59 of 135 visitors liked the old layout of the website. Company claims that new layout did not improve the visitor's liking about the website.

By using the critical value $Z < -Z_{0.05} = -1.645$, verify the company's claim:

$H_o : P_1 - P_2 \geq 0$

$H_1 : P_1 - P_2 < 0$

Answer (Please click here to Add Answer)



Made by: Waqar Siddhu

Start Time: 9:48 AM

52

Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The value of χ^2 can never be :

Answer (Please select your correct option)

- Zero
- Less than 1
- Greater than 1
- Negative

Correct answer solved by HADI
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Cell : 03228043306

correct answer
solved by hadi

Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Analysis of variance is a procedure that enables us to test the equality of several:

Answer (Please select your correct option)

- Variances
- Means

correct answer
solved by hadi

- Proportions
- Groups

Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When two coins are tossed the probability of at most one head is:

Answer (Please select your correct option)

- 1/4
- 2/4
- 3/4

correct answer
solved by hadi

- 4/4

Question No : 5 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

The critical region for $H_1 : \mu < \mu_0$ *when* $\alpha = 0.05$ is:

Answer (Please select your correct option)

- $z > z_{0.05}$ — **correct answer solved by hadi**

Correct answer solved by HADI
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- $|z| > z_{0.10}$
- $z < -z_{0.05}$
- $|z| > z_{0.05}$

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Question No : 6 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

A discrete probability function f(x) is always:

Answer (Please select your correct option)

- Non-negative
- Negative
- One — **correct answer solved by hadi**
- Zero

Made by: Waqar Siddhu

Question No : 7 of 52 — **Marks: 1 (Budgeted Time 1 Min)**

When f(x) is continuous probability function, then P(X = 1) is:

Answer (Please select your correct option)

- 1
- ∞
- $-\infty$
- 0 — **correct answer solved by hadi**

Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A random variable can be generated:

Answer (Please select your correct option)

- Manually
- Mechanically
- Manually & Mechanically
- Mathematically

Correct answer solved by HADI
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Cell : 03228043306

correct answer solved by hadi

Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The curve of the F- distribution depends upon:

Answer (Please select your correct option)

- Degrees of freedom

correct answer solved by hadi

- Standard deviation
- Mean
- Variance

Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

- 0.6
- 0.8

correct answer solved by hadi

- 0.2
- 0.4

Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The mode value from raw data can be obtained by the help of

Answer (Please select your correct option)

- Dot plot
- Stem and leaf plot
- Bar chart
- None of these

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In normal distribution, the quartile deviation Q.D = ... :

Answer (Please select your correct option)

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

correct answer
solved by hadi

Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

Answer (Please select your correct option)

- It is a sample statistic.
- It is a population parameter.
- It is a margin of error.
- It is a standard error.

Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A student solved 25 questions from first 50 questions of a book . The probability that he will solve the remaining all questions is:

Answer (Please select your correct option)

- 0.25
- 0.5

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correct answer solved by hadi

- 1
- 0

Question No : 15 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

- Geometric Mean
- Harmonic Mean

correct answer solved by hadi

- Mode
- Median

Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A∩B) =……..

Answer (Please select your correct option)

- 0.65
- 0.1

correct answer solved by hadi

- 0.50
- 0.15

Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The total number of samples when sampling is done with replacement is equal to:

Answer (Please select your correct option)

- N^n

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correct answer
solved by hadi

- C_n^N
- $\frac{N-n}{N-1}$
- 1

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is correct option to find $P(X+Y \leq 1)$?

Answer (Please select your correct option)

- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

correct answer
solved by hadi

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

The conditional probability function $f(x \mid 1) =$ _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$

correct answer
solved by hadi

- $\frac{f(x,1)}{h(x)}$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

In interval estimation, we always get:

Answer (Please select your correct option)

- Range of values

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correct answer
solved by hadi

- Zero
- Single value
- Two values

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is an unbiased estimator of population σ^2 ?

Answer (Please select your correct option)

- $\frac{\sum(x-\bar{x})^2}{n}$
- $\frac{\sum(x-\bar{x})^2}{n-2}$
- $\frac{\sum(x-\bar{x})^2}{n(n-1)}$
- $\frac{\sum(x-\bar{x})^2}{n-1}$

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

How many parameter(s) are in t-distribution?

Answer (Please select your correct option)

- 0
- 1

correct answer
solved by hadi

- 2
- 3

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

- Type I error
- Type II error

correct answer
solved by hadi

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- Correct decision
- No information regarding this

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

After an interval is constructed, then what is the probability of occurrence of the parameter in it?

Answer (Please select your correct option)

- Zero
- One
- Fifty

correct answer
solved by hadi

- Either one or zero

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

Answer (Please select your correct option)

- Z-statistic
- T-statistic
- χ^2-statistics

correct answer
solved by hadi

- F-statistics

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k

correct answer
solved by hadi

Correct answer solved by HADI
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- 1/k
- k

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

- Positively skewed

correct answer
solved by hadi

- Negatively skewed
- Uniformly distributed
- Normally distributed

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

- Degree of freedom

correct answer
solved by hadi

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- Critical value
- Frequency of data
- Probability

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

If X and Y are random variables, then $E(X-Y)$ is equal to:

Answer (Please select your correct option)

- $E(X)+E(Y)$
- $E(X)-E(Y)$ correct answer solved by hadi
- $X-E(Y)$
- $E(X)-Y$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In the regression line Y= a + bX, the non-random variable is:

Answer (Please select your correct option)

- Y correct answer solved by hadi but check
- X
- Both X and Y
- Neither X nor Y

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The parameters of the binomial distribution b(x; n, p) are:

Answer (Please select your correct option)

- x & n
- x & p
- n & p correct answer solved by hadi

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- x, n & p

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is true for the binomial distribution b(x; n, p):

Answer (Please select your correct option)

- mean > variance

correct answer
solved by hadi

- mean < variance
- mean = variance
- mean= standard deviation

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Hypergeometric probability distribution has :

Answer (Please select your correct option)

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

correct answer
solved by hadi

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

The variance of the hypergeometric probability distribution is:

Answer (Please select your correct option)

- $\sigma^2 = n\frac{k}{N}$
- $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$,

correct answer
solved by hadi

Correct answer solved by HADI
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- $\sigma^2 = npq$
- $\sigma^2 = N\frac{n}{K}$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Tabulation is the process of arranging data into:

Answer (Please select your correct option)

- Different classes
- Rows
- Columns
- Rows and Columns

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A simple bar chart consists of:

Answer (Please select your correct option)

- Vertical bars
- Horizontal bars
- Vertical or horizontal bars
- Multiple bars

correct answer
solved by hadi

Correct answer solved by HADI
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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is the formula to calculate the approximate value of class interval?

Answer (Please select your correct option)

- Minimum value/Range
- Maximum value/No. of classes
- No. of classes/Range
- Range/No. of classes

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is equal to the 2nd quartile:

Answer (Please select your correct option)

- P_{33}
- D_3
- Median — correct answer solved by hadi
- Mode

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

For the independent events A and B if P (A) = 0.15, P (B) =0.50 then P (A∩B) =.....

Answer (Please select your correct option)

- 0.50
- 0.075 — correct answer solved by hadi

Correct answer solved by HADI
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- 0.125
- 0.060

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following measure is not based on all the observations?

Answer (Please select your correct option)

- Arithmetic Mean
- Geometric Mean
- Harmonic Mean
- Mode — correct answer solved by hadi

Correct answer solved by HADI
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Made by: Waqar Siddhu

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Is there any difference between distribution function and cumulative distribution function of discrete r.v. X?

Answer (Please click here to Add Answer)

VuAnswers.com

The distribution function of a random variable X, denoted by F(x), is defined by F(x) = P(X < x).
The function F(x) gives the probability of the event that X takes a value LESS THAN OR EQUAL TO a specified value x. The distribution function is abbreviated to *d.f.* and is also called the *cumulative distribution function (cdf)* as it is the cumulative probability function of the random variable X from the smallest value upto a *specific* value x.
Let us illustrate this concept with the help of the same example that we have been considering --- that of the probability distribution of the sum of the dots when two fair dice are thrown.

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by sampling distribution of sample proportion?

Answer (Please click here to Add Answer)

VuAnswers.com

The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution.

The sample proportion has different values in different samples.

Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

If an automobile is driven on the average no more than 16000 Km per year then formulate the null and alternative hypothesis.

Answer (Please click here to Add Answer)

VuAnswers.com

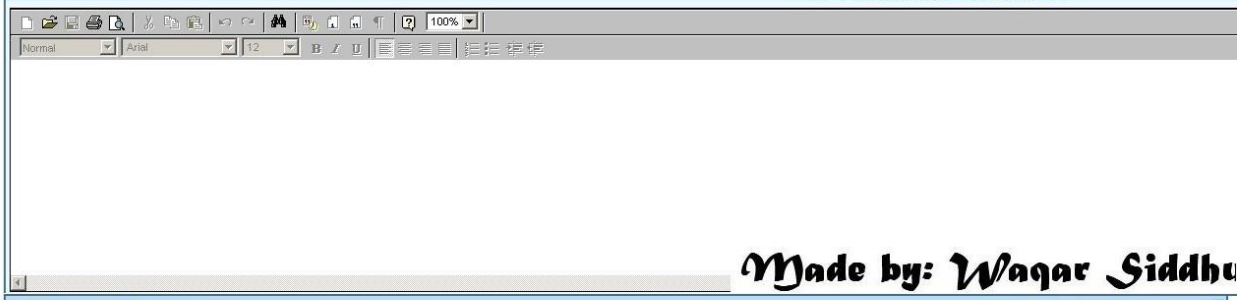
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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

While applying the chi-square goodness of fit test, if an expected frequency in any category is less than 5, then what we do to solve this problem.

Answer (Please click here to Add Answer)

VuAnswers.com



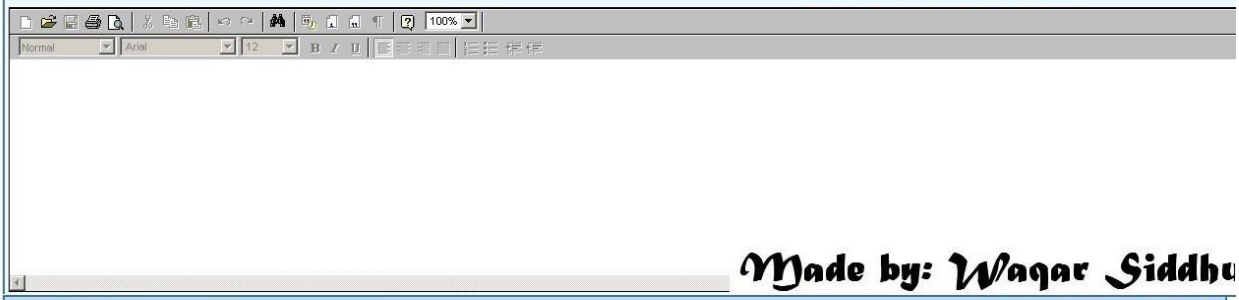
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Write a short note on Quota Sampling.

Answer (Please click here to Add Answer)

VuAnswers.com



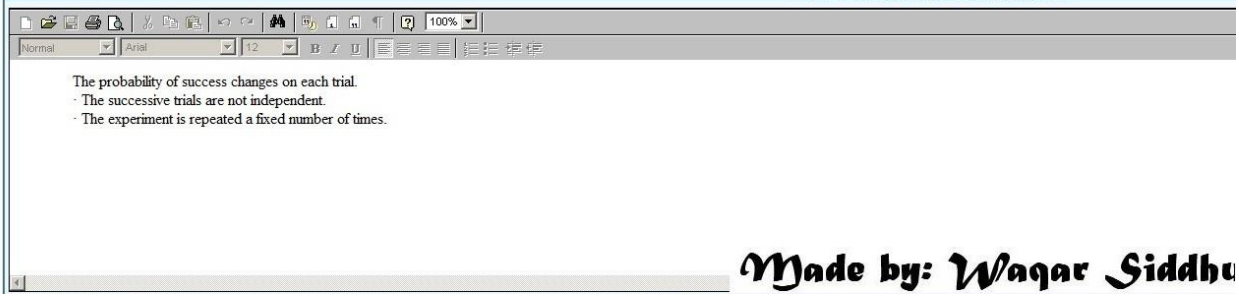
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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down the properties of hypergeometric distribution.

Answer (Please click here to Add Answer)

VuAnswers.com



The probability of success changes on each trial.
- The successive trials are not independent.
- The experiment is repeated a fixed number of times.

Made by: Waqar Siddhu

Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down critical region for the following hypothesis at 5% level of significance.

$H_0 : \mu = 75$

$H_1 : \mu < 75$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

For the following table we want to test the independence of smoking pattern and marital status. What will be the degrees of freedom for the chi-square test of independence?

Marital Status	Smoking Pattern			Total
	Total self-restraint	Only at times	Regular Smoker	
Single	67	213	74	354
Married	411	633	129	1173
Widowed	85	51	7	143
Divorced	27	60	15	102
Total	590	957	225	1772

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Answer (Please click here to Add Answer)

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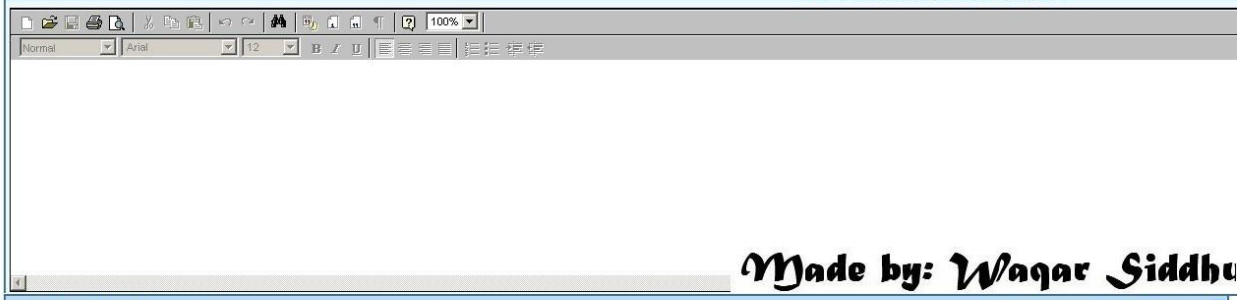
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Answer the following question using the given stem and leaf plot of the ages of the people who attended a musical event.

Stem	Leaf
1	2 2 3 3 4 4 5 5 5 5 5 5 6 6 6 6 6 6 7 7 7 7 7 8 8 9 9
2	3 3 4 8
3	6

Answer (Please click here to Add Answer)

VuAnswers.com



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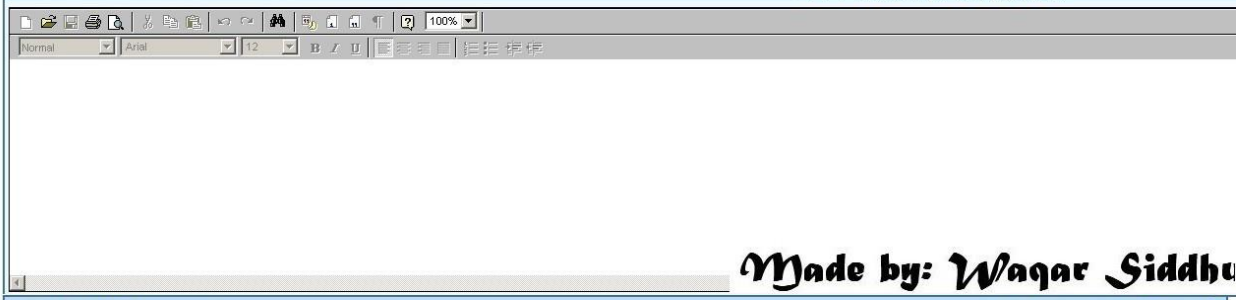
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

2	3 3 4 8
3	6
4	5 8

a) How many people attended the event?
b) What is the age of youngest attendee?

Answer (Please click here to Add Answer)

VuAnswers.com



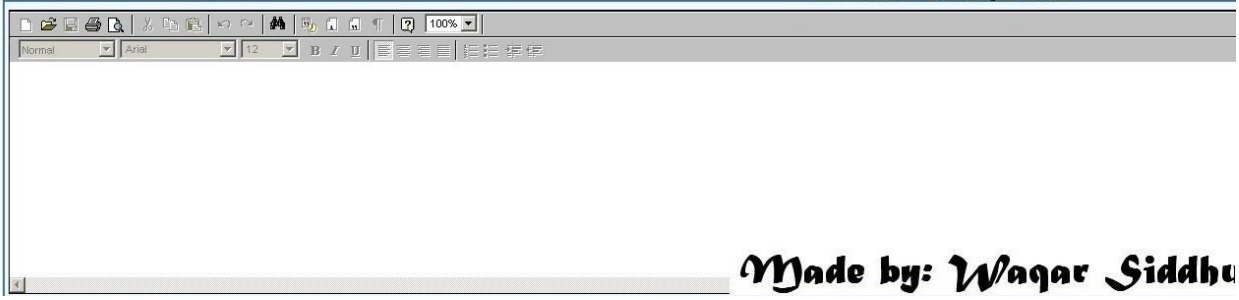
Made by: Waqar Siddhu

Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

a) How many people attended the event?
b) What is the age of youngest attendee?
c) What was the age of oldest person attending event?
d) Which age group was more widely represented?
e) How many people were above 30?

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

If $\bar{X}$ is normally distributed with $\mu_{\bar{x}} = \mu = 24000$ and $\sigma_{\bar{x}} = 412.20$

then find out the $P(\bar{X} > 24500)$.

Answer (Please click here to Add Answer)

VuAnswers.com

Made by: Waqar Siddhu

Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

If we have $\bar{X} = 15$ and $\sum_{i=1}^{10}(X_i - \bar{X})^2 = 56$

Then, find the 90% confidence interval for population variance (σ^2).

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

A research worker wishes to estimate the mean of a population using a sample sufficiently large that the probability will be 0.95 that the sample mean will not differ from the true mean by more than 25 percent of the standard deviation. How large a sample should be taken?

Answer (Please click here to Add Answer)

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Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Var(4X + 5) =_____

Answer (Please select your correct option)

- 16 Var (X)
- 16 Var (X) + 5
- 4 Var (X) + 5

correct answer
solved by hadi

Correct answer solved by HADI
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- 12 Var (X)

Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When f(x) is continuous probability function, then P(X = 1) is:

Answer (Please select your correct option)

- 1
- ∞
- −∞
- 0

correct answer
solved by hadi

Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The hyper geometric random variable is a(an):

Answer (Please select your correct option)

- Continuous variable
- Discrete variable

correct answer
solved by hadi

- Undefined
- Independent variable

Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

- n-1 — correct answer solved by hadi

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- n+1
- n-2
- n+2

Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

- $H_0 : \mu = 100$ — correct answer solved by hadi
- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

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Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

- 0.6
- 0.8 — correct answer solved by hadi
- 0.2
- 0.4

Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In normal distribution β_1 = ... :

Answer (Please select your correct option)

- 0 — correct answer solved by hadi

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- 1
- 2
- 3

Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

- Repeated sampling
- Small sample
- Large sample — correct answer solved by hadi
- Large population

Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

Answer (Please select your correct option)

- It is a sample statistic.
- It is a population parameter.
- It is a margin of error.
- It is a standard error. — correct answer solved by hadi — but not sure

Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

- A population distribution
- A sample distribution
- A sampling distribution

correct answer
solved by hadi

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- A parameter distribution

Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The conditional probability function $f(x \mid 1) =$ _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$

correct answer
solved by hadi

- $\frac{f(x,1)}{h(x)}$

Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If f (x, y) is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

- $\int_{-\infty}^{\infty} f(x,y)\,dx$

correct answer
solved by hadi

Correct answer solved by HADI
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- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$

Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

- Hypothesis
- Lower limit
- Confidence interval

not sure

- Probability limit

Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

How many parameter(s) are in t-distribution?

Answer (Please select your correct option)

- 0
- 1

correct answer
solved by hadi

- 2
- 3

Question No : 15 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision

correct answer
solved by hadi

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- No information regarding this

Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

- $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 2.58\sigma_{\bar{x}}$ — correct answer solved by hadi
- $\mu_{\bar{x}} \pm \sigma_{\bar{x}}$

Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$ — correct answer solved by hadi
- $P \geq 0.48$

Question No : 18 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k — correct answer solved by hadi — Correct answer solved by HADI Email : usmanraj20@gmail.com Cell : 03228043306
- 1/k
- k

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Question No : 19 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Test Statistics χ^2 is equal to:

Answer (Please select your correct option)

- $\frac{n^2S^2}{\sigma_0^2}$
- $\frac{nS^2}{\sigma_0^2}$ correct answer
  solved by hadi
- $\frac{S^2}{2\sigma_0^2}$
- $\frac{n^2S^2}{2\sigma_0^2}$

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Question No : 20 of 52 **Marks: 1 (Budgeted Time 1 Min)**

F- distribution is .....

Answer (Please select your correct option)

- Positively skewed
  correct answer
  solved by hadi
- Negatively skewed
- Normal
- Symmetrical

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Question No : 21 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The Chi- Square distribution is continuous distribution ranging from:

Answer (Please select your correct option)

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- $0 \leq \chi^2 \leq \infty$
  correct answer
  solved by hadi

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Question No : 22 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In Statistics, we have MSE which is abbreviation of:

Answer (Please select your correct option)

- Mean square error

correct answer
solved by hadi

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- Measured square error
- Medical screening exam
- Major sampling error

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The LSD test is applied when the null hypothesis is:

Answer (Please select your correct option)

- Rejected

correct answer
solved by hadi

- Accepted
- Finalized
- Acknowledged

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Question No : 24 of 52 **Marks: 1 (Budgeted Time 1 Min)**

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In a binomial experiment the total number of trials are:

Answer (Please select your correct option)

- Fixed in advance

correct answer
solved by hadi

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- Changeable according to situation
- Unpredictable
- Not independent

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Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

What is the probability of drawing a red-queen card from a well shuffled pack of 52 playing cards?

Answer (Please select your correct option)

- 4/52
- 2/52

correct answer
solved by hadi

- 13/52
- 26/52

Made by: Waqar Siddhu

Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The probability of drawing a spade card is:

Answer (Please select your correct option)

- 1/52
- 4/52
- 13/52

correct answer
solved by hadi

- 26/52

Made by: Waqar Siddhu

Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is true for the binomial distribution b(x; n, p):

Answer (Please select your correct option)

- mean > variance — correct answer solved by hadi

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- mean < variance
- mean = variance
- mean= standard deviation

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Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When f(x) is continuous probability function, then P(X = 2) is:

Answer (Please select your correct option)

- 1
- 0.5
- 0 — correct answer solved by hadi
- 0.25

Made by: Waqar Siddhu

Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

- Unbaised
- Sufficient
- Efficient — correct answer solved by hadi
- Consistent

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Question No : 31 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In construction of a histogram, what would be taken along X-axis?

Answer (Please select your correct option)

- Mid points
- Class limits
- Class interval
- Class boundaries

correct answer
solved by hadi

Made by: Waqar Siddhu

Question No : 32 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

Answer (Please select your correct option)

- Ogive
- Frequency polygon
- Frequency curve

correct answer
solved by hadi

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- Historigram

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Question No : 33 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Tabulation is the process of arranging data into:

Answer (Please select your correct option)

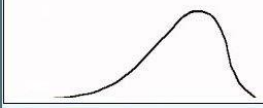
- Different classes
- Rows
- Columns
- Rows and Columns

correct answer
solved by hadi

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Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

- Negatively skewed curve

  correct answer
  solved by hadi
- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If A = {1, 2, 3, 4, 5, 10} and B = {1, 3, 5} then B⊂A means:

Answer (Please select your correct option)

- A is less than B
- A is contained in B
- B is contained in A.

  correct answer
  solved by hadi

  Correct answer solved by HADI
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  Cell : 03228043306
- B is less than A

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

What is the mode in the word STATISTICS:

Answer (Please select your correct option)

- S
- T
- I
- S and T

  correct answer
  solved by hadi

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Question No : 37 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In a symmetrical distribution, if $Q_1 = 20$, $Q_2 = 30$ *then* Q_3 *is* :

Answer (Please select your correct option)

- 50
- 40

correct answer
solved by hadi

- 30
- 10

Made by: Waqar Siddhu

Question No : 38 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following is not included in measures of central tendency:

Answer (Please select your correct option)

- Quartile Deviation

correct answer
solved by hadi

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- Harmonic Mean
- Geometric Mean
- Arithmetic Mean

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Question No : 39 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The sum of deviations from mean is:

Answer (Please select your correct option)

- Maximum
- Minimum
- Zero

correct answer
solved by hadi

- Undefined

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Question No : 40 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Mean deviation is always:

Answer (Please select your correct option)

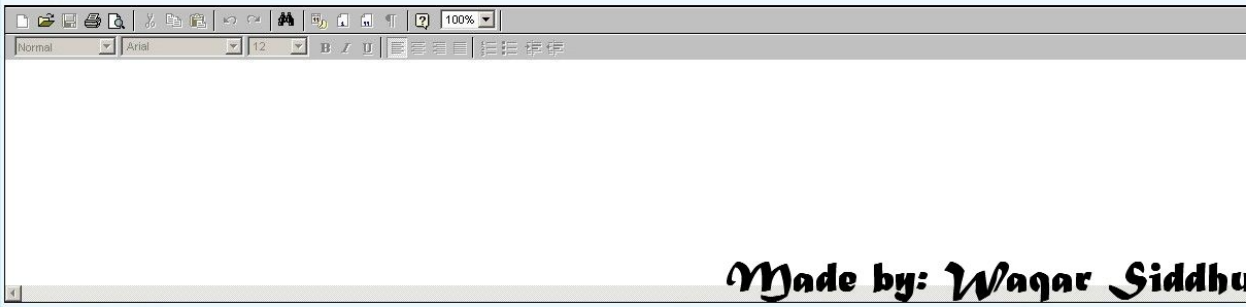
- Less than Standard Deviation
- Greater than Standard Deviation
- Greater or equal to Standard Deviation
- Less or equal to Standard Deviation

Correct answer solved by HADI
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correct answer
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Question No : 41 of 52 **Marks: 2 (Budgeted Time 4 Min)**

In how many branches, estimation can be divided? Also write down the names of the branches.

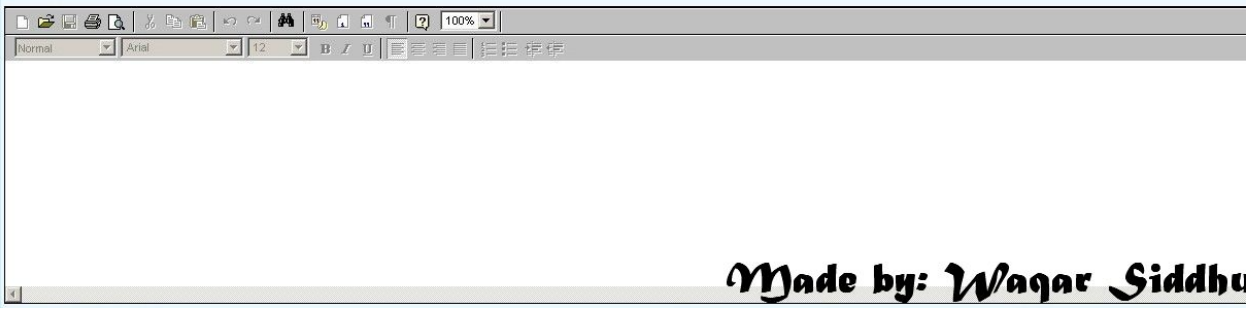
Answer (Please click here to Add Answer)



Question No : 42 of 52 **Marks: 2 (Budgeted Time 4 Min)**

A Sample has following values.
101, 99, 111, 199, 150, 140
Find out the maximum likelihood estimate of population mean.

Answer (Please click here to Add Answer)



Question No : 43 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Write down critical region for the following hypothesis.

$H_0 : \sigma^2 = 20$

$H_1 : \sigma^2 < 20$

Where, $\alpha = 0.01$ and n=10

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 44 of 52 **Marks: 2 (Budgeted Time 4 Min)**

What is the concept of extrapolation in regression analysis?

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 45 of 52 **Marks: 3 (Budgeted Time 6 Min)**

If Z is a standard normal random variable with mean 0 and variance 1, then find the Lower quartile.

Answer (Please click here to Add Answer)

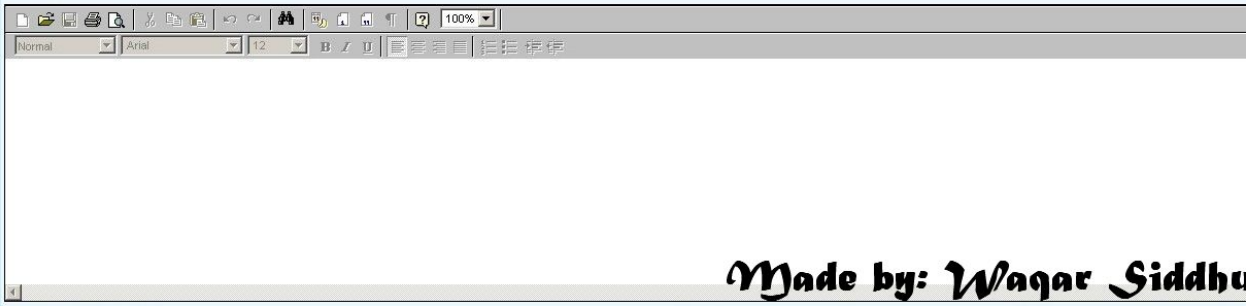
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Question No : 46 of 52 **Marks: 3 (Budgeted Time 6 Min)**

For the following table we want to test the independence of smoking pattern and marital status. What will be the degrees of freedom for the chi-square test of independence?

Marital Status	Smoking Pattern			Total
	Total self-restraint	Only at times	Regular Smoker	
Single	67	213	74	354

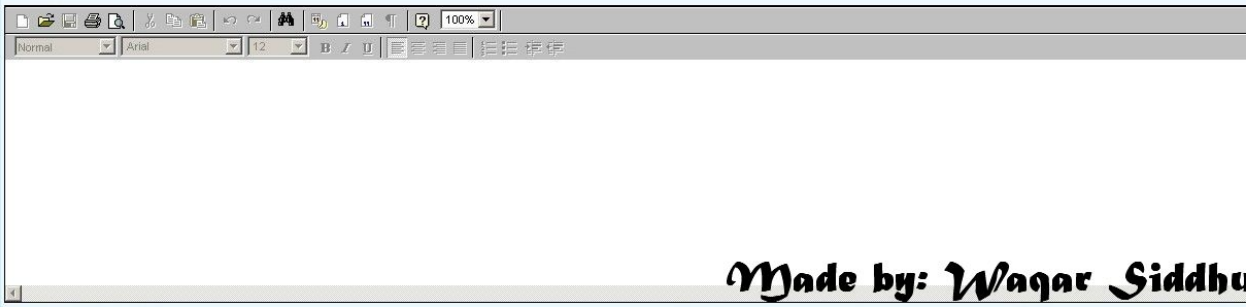
Answer (Please click here to Add Answer)



Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Find the proportion for the X_1 and X_2, Where X_1 =300 with sample size= n_1 = 400 and X_2 = 200 with sample size = n_2 =300.

Answer (Please click here to Add Answer)

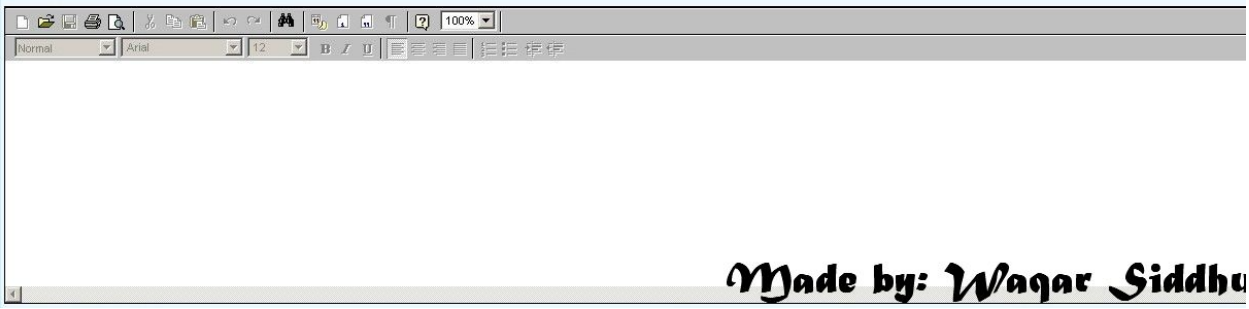


Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Calculate class boundaries from the following data.

Age Group	f
20-29	2
30-39	4
40-49	5

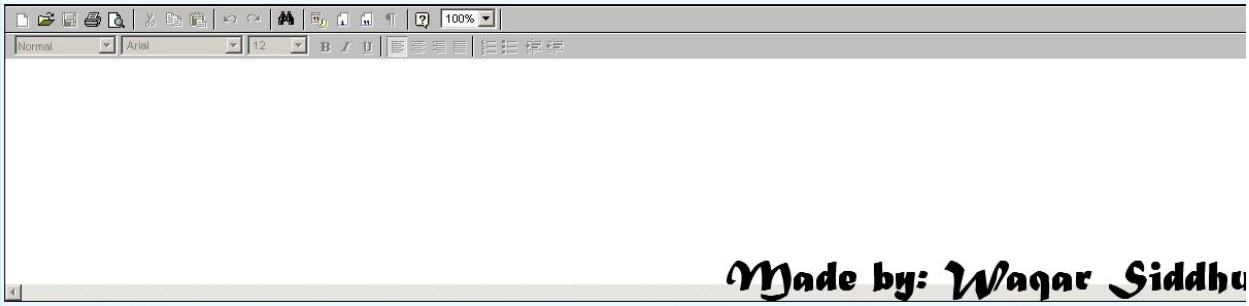
Answer (Please click here to Add Answer)



Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A certain type of storage battery lasts on the average 3.0 years, with a standard deviation of 0.5 year. Assuming that the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years.

Answer (Please click here to Add Answer)

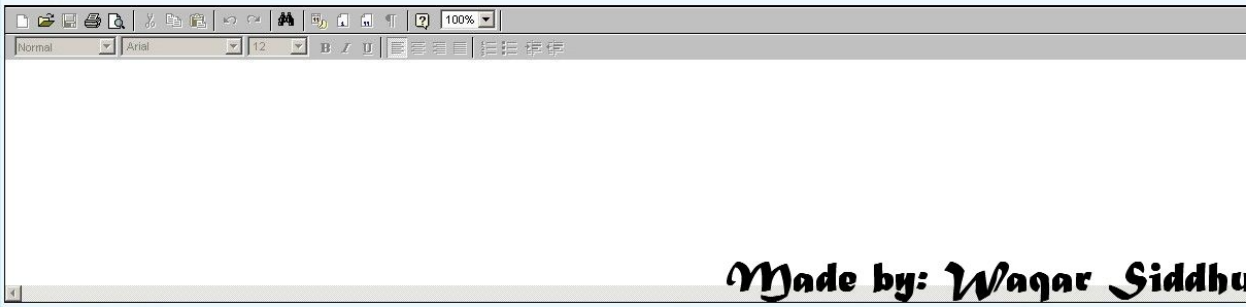


Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

For the Given information, what is your conclusion in testing the indicated null hypothesis?

$n_1 = 13, n_2 = 41, s_1^2 = 6.3, s_2^2 = 15.6$, $H_0 : \sigma_1^2 = \sigma_2^2$ and $H_1 : \sigma_2^2 > \sigma_1^2$

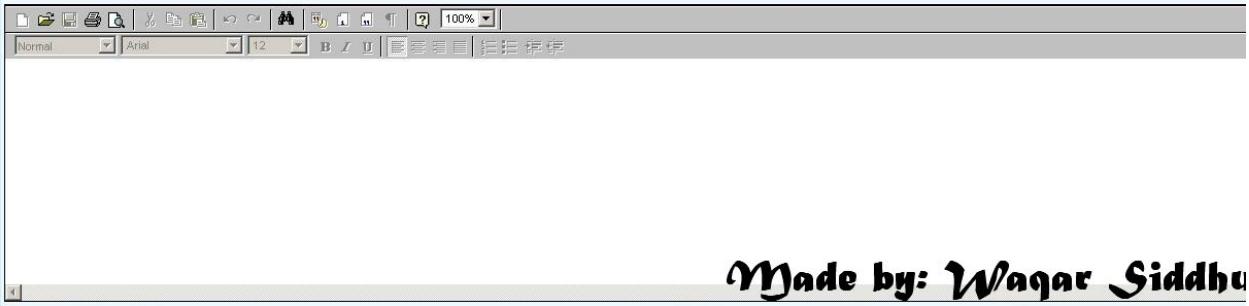
Answer (Please click here to Add Answer)



Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

In a random sample of 200 persons having their lunch at the University cafeteria on meatless day it was observed that 30 percent preferred vegetable dishes.Find 95% confidence interval for p.

Answer (Please click here to Add Answer)



Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Answer the following questions:

i. Which method has been used for colleting the data in the following case?
"An investigator collects the information personally from the individuals concerned".

i. Find the sampling error when the population mean is 15.785 and sample mean is 15.6.

ii. Write down any two types of nonrandom sampling

iii. What is the singular of strata?

Answer (Please click here to Add Answer)

